Planet Hollywood: Do Film Production Incentives Attract International Production? A Gravity Model Approach

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20 April 2026

Abstract

This paper builds a gravity model of international film production flows and examines the effectiveness of national film production incentive schemes in attracting international film productions. I construct bilateral trade data using a dataset sourced from the Internet Movie Database (IMDb) covering the 600 top-grossing films per year from 1990 to 2023. I find that the gravity framework fits bilateral film production flows well, but the determinants differ in notable ways from those documented for goods trade. Cross-sectional estimates indicate that countries with active incentive schemes receive approximately 18% more inward production, but this reduces to roughly $9 - 11\%$ under country-pair fixed effects. Early adopters (pre-2005) attract substantially more production than late adopters, while incentive effects show no evidence of compounding over time.

Word Count: XXX

1 Introduction

1.1 Motivation

The international film industry was worth an estimated $112 billion in 2025 [Grand View Research, 2026], and has become increasingly globalized in recent decades with the rise of "runaway productions", whereby films originating in one country are produced in another [Morawetz et al., 2007]. The perceived economic value of hosting film productions such as employment and tourism spillovers, as well as the cultural prestige of a domestic film industry, has led to a large number of countries implementing incentives to attract international productions. The effectiveness and economic benefits of these schemes have been hotly constested given the large fiscal costs involved [Thom, 2018].

1.2 Research Question and Approach

This paper asks how well a gravity model predicts international film production flows, whether film production incentives are effective at attracting international productions, and whether any observed correlation between incentive adoption and inward production reflects selection or true causation. The analysis applies a gravity model framework to bilateral film production flows, treating filming-location choice as a form of creative services trade between an origin country (which finances production) and a destination country (where physical filming occurs). I use filming-location data and budget data extracted from the Internet Movie Database (IMDb) covering the 600 top-grossing films per year from 1990 to 2023 as my primary dataset to construct bilateral trade flows. I also utilise the shorter OECD-WTO Balanced Trade in Services (BaTIS) data on bilateral audiovisual services exports from 2005 to 2023 as secondary robustness check.

I use cross-sectional Poisson Pseudo-Maximum Likelihood (PPML) estimates with importer, exporter, and year fixed effects to establish the unconditional association between incentive adoption and inward production. Country-pair fixed effects specifications are then used to identify incentive effects exclusively from within-pair temporal variation, this being the closest approximation available to a causal effect.

1.3 Headline Findings

Three main findings emerge. First, the gravity framework fits bilateral film production flows well, but the determinants differ in notable ways from those documented for goods trade. Distance elasticities are substantially smaller, common language effects are larger, and colonial ties and regional trade agreements are associated with less production rather than more. However, these patterns are broadly consistent with the language-intensive and culture-specific nature of film as a product. Second, incentive schemes show strong evidence of attracting international production, and weaker evidence of re-shoring film production. Cross-sectional estimates imply that incentive schemes drive approximately 18% more international film productions. This reduces to roughly $9-11\%$ once country-pair fixed effects are applied but is robust across subsamples. The introduction of an incentive scheme in one country also appears to reduce the number of films made in another by 9% within country pairs, although this is not corroborated by the budget data or BaTIS robustness check. Third, the timing of adoption matters but the duration of operation does not. Early adopters (pre-2005) attract 26% more films and 45% more production spending, while late adopters show a smaller and less precisely estimated effect. Incentives show no evidence of a compounding "cluster" effect.

1.4 Contribution

This paper makes two contributions to the literature on film production incentives. First, it is the first study to apply the gravity model of trade to international film production flows. The existing literature on incentive effectiveness focuses almost exclusively on the U.S. state level where many determinants of production location such as language and geographic distance do not apply. Applying the gravity framework at the international level allows the effects of incentives to be estimated against larger set of factors that shape production decisions. Second, the paper offers an explicit decomposition of cross-sectional and within-pair incentive effects. The gap between the PPML and pair fixed effects estimates provides a direct quantification of the selection component of the observed correlation between incentive adoption and inward production. This effect is estimated

in parallel across two independent datasets to add robustness to the findings.

1.5 Roadmap

The remainder of the paper is organised as follows. Section 2 reviews the existing literature. Section 3 describes the construction of bilateral film production flows from the IMDb data and the assembly of standard gravity controls. Section 4 sets out the methodology. Section 5 presents the estimation results and the cross-dataset validation against BaTIS. Section 6 discusses limitations of the analysis. Section 7 concludes.

2 Literature Review

2.1 International Film Industry

A variety of factors influence the distribution of international film production including labour costs, government incentives, co-production treaties, and the existence of pre-existing production networks [Morawetz et al., 2007, Owens and Rennhoff, 2020, Mitric, 2018]. In recent decades, a growing number of national and subnational governments have introduced schemes to attract international film productions [Christopherson and Rightor, 2010]. These incentive schemes take different forms including cash rebates and tax credits, and vary substantially across jurisdictions with respect to criteria such as credit rates, eligibility criteria, and cultural content requirements [Olsberg SPI, 2025].

The net economic impact of production incentives is debatable. The fiscal cost of film production subsidies often outstrips the tax revenue they generate [Luther, 2010, Tannenwald, 2010], and several studies have found evidence of "crowding out" where film production incentives merely draw resources from other sectors of the economy [Calcagno and Thompson, 2004]. Conversely, many industry-commissioned studies have found substantial positive returns from incentive schemes [Ernst & Young, 2009, Olsberg SPI, 2023]. However, the methodological basis of these studies has been questioned [Button, 2021].

A related but limited set of literature sets aside the issue of net welfare effects and examines whether incentives succeed at attracting film productions. These studies focus

exclusively on incentives at the U.S. state level. This is due to the large number of state-level schemes available and their similar characteristics reducing potential confounding factors. Some of these studies find meaningful evidence that incentive schemes can attract film productions. Button [2021] employs synthetic control and difference-in-differences methods to evaluate the programmes in Louisiana and New Mexico (two early adopters of generous state film incentives) and finds that incentives lead to a statistically significant increase in the number of films produced. Workman [2021] exploits the fact that California's Film and Television Tax Credit Program allocates tax incentives by lottery, creating quasi-random variation in which productions receive incentives and allowing him to estimate causal effects more cleanly than other studies. He finds that the offer of an incentive meaningfully increased the number of films produced in the state.

Other studies are more mixed. Owens and Rennhoff [2020] estimate a discrete choice model of filming location decisions for productions released between 1999 and 2013. They find significant heterogeneity in responsiveness: mid-sized studios respond to all forms of incentives, major studios respond only to refundable and transferable tax credits, and independent productions exhibit no statistically significant sensitivity to incentives. Finally, Button [2019] uses a panel difference-in-differences design comparing U.S. states before and after they adopt film incentives against similar states that did not adopt over the same period. He finds an increase in the production of TV series but no meaningful effect on feature film production.

2.2 The Gravity Model of Trade

The gravity model of trade predicts that bilateral trade between two countries is proportional to the product of their economic sizes and inversely related to the geographic distance between them [Tinbergen, 1962]. It has been shown to be derivable from a wide range of popular trade models [Eaton and Kortum, 2002, Krugman, 1980, Chaney, 2008], and has been extensively used to model trade flows [Head and Mayer, 2014]. When estimating gravity models of international trade, it has become standard to include dummy variables for each exporting and importing country in the sample to control for all un-

observable country-specific factors that affect a country's overall propensity to trade [Anderson and van Wincoop, 2003].

The gravity model has been extensively applied to goods trade but employed less frequently to trade in services. Santanna et al. [2023] attribute this to three factors: well-known limitations in bilateral services trade data, the comparatively recent emergence of services as a significant component of international trade, and a widespread assumption that insights from the goods trade literature transfer directly to services. However, existing literature suggests that the gravity framework is well suited to modelling services flows. Kimura and Lee [2006] estimate gravity equations for bilateral services trade between ten OECD member countries and a broader sample of trading partners. Their principal finding is that services trade is better predicted by the gravity specification than goods trade, and that there exists a complementary relationship between goods exports and services imports.

Walsh [2006] applies the gravity model to aggregate services trade and finds that the gravity model fits services trade in a manner broadly comparable to goods. However, he finds that GDP per capita and common language emerge as the most significant determinants of services trade, while geographic distance is generally insignificant. This is consistent with the observation that many services do not require physical shipment and are therefore less sensitive to transport costs but notably diverges from the goods gravity literature which finds distance to be the dominant friction in international trade. Thai [2006], analysing Vietnam–EU bilateral services flows, confirms these patterns and documents the significance of GDP and common language as determinants.

3 Data

3.1 Bilateral Trade Flow Construction

The primary dataset used for this analysis is sourced from the Internet Movie Database (IMDb) and contains the 600 top-grossing films per year from 1990–2023, yielding approximately 20,400 films. Each film contains metadata such as their name, countries of origin,

primary filming location, budget, and release year. However, coverage of key variables differs across the panel. While countries of origin are available for most films, filming location data is available for approximately 70% of films. Furthermore, a lower proportion of non-English-language productions have budgets in USD, so the budget data under-represents non-English-language productions. This limitation is addressed through the use of subsamples for the Fixed Effects specifications that exclude the English-speaking Anglophone countries (United States, Canada, United Kingdom, Australia, and New Zealand).

Filming locations are extracted by parsing IMDb address strings, taking the final comma-separated element as the country name (e.g., "Winnetka, Illinois, USA" yields "United States"). A comprehensive mapping file harmonises over 160 variant country names and sub-national entities to standardized ISO country codes. For each film with a valid filming country and at least one country of origin, I construct bilateral flows from each origin country to the filming country. Only films which contain non-indentical countries of origin and filming locations are retained, thereby excluding purely domestic productions. This leaves approximately 7,579 films.

Two dependent variables are constructed at the country-pair-year level. The primary dependent variable is the number of films produced per year, while the secondary dependent variable is the total production budgets in constant 2010 USD, which is available for approximately 47% of films. For films with multiple countries of origin, each origin is paired with this single filming country to generate a separate bilateral flow. When a film has multiple countries of origin, the full budget is divided equally across origin countries.

As a robustness check, I also test using the OECD–WTO Balanced Trade in Services (BaTIS) 'Audiovisual and related services data which captures bilateral exports of film, television, radio, and related production services from 2005 to 2023. While the BaTIS data captures non-film related audiovisual services trade, it also captures other filmmaking related activity such as post-production, visual effects, and animation work. While this activity is not being directly captured by the IMDb data (which relies on physical filming locations), it can also be expected to be closely intertwined with physical filming

location decisions and therefore respond to similar factors such as film incentives, making it an appropriate robustness check. As with film budgets, these values are deflated to constant 2010 USD.

3.2 Gravity Variables

Standard bilateral gravity variables are sourced from the CEPII GeoDist database and include bilateral distance (population-weighted, in logs), contiguity, common official language, colonial relationship post-1945, and regional trade agreements. Where the CEPII data coverage ends at 2020, the most recent available year is forward-filled to 2023 under the assumption that these predominantly time-invariant variables change minimally over this extension period.

The Economic Freedom of the World (EFW) index data is sourced from the Fraser Institute, both measured for importer and exporter. The EFW index is published annually only from 2000 onward; prior to 2000, it is published at five-year intervals. To obtain a complete annual panel for the 1990–2023 period, linear interpolation is applied within each country's time series for the missing years, as is standard in the literature using EFW data.

Annual GDP and inflation data is sourced from the World Bank. The GDP data is deflated to constant 2015 US dollars, with the natural logarithm of GDP being used in the gravity specification.

3.3 Film Incentive Data

Data on national film production incentive schemes for 63 countries is sourced from the Olsberg SPI Global Incentives Index 2025 [Olsberg SPI, 2025]. For each country, the dataset records the year of incentive introduction, the headline rebate or credit rate (as a percentage of qualifying production expenditure), and the incentive type. Incentive types follow the three-category classification used by Olsberg SPI: **rebates**, which return a percentage of qualifying production expenditure as cash grants; **tax credits**, which offer a reduction against corporate tax owed; and **tax shelters**, which are funded by investors

seeking tax relief in exchange for financing production. Each country is assigned a single category corresponding to its primary national scheme and a rate as a % of production costs. Of the 57 countries with an active national incentive scheme in the dataset, 48 operate a rebate, 6 operate a tax credit, and 3 operate a tax shelter. In order to fit within the country-apir framework, subnational incentives are excluded from the analysis and countries with subnational schemes (such as the U.S) but lack a national scheme are categorised as not having an incentive scheme. Where Olsberg reports tiered or variable headline rates (e.g., rates that vary by expenditure band or include uplifts for local hiring), the midpoint across tiers is used.

4 Methodology

4.1 The Gravity Model Framework

Following Kimura and Lee [2006], the standard gravity equation drawn from theory takes the following form:

$$T_{ij} = G_i^{\beta_1} \cdot G_j^{\beta_2} \cdot D_{ij}^{\beta_3} \cdot R_i^{\beta_4} \cdot R_j^{\beta_5} \cdot E_{ij} \tag{1}$$

where T_{ij} denotes bilateral trade flows between country i and country j; G_i and G_j represent the economic mass of each country (proxied by GDP); D_{ij} is the geographical distance between countries i and j; R_i and R_j capture the relative distance (remoteness) of each country from the rest of the world; and E_{ij} is a multiplicative error term. Taking logarithms yields the standard log-linear estimating equation.

4.2 Model Specifications

Ten models are estimated, building sequentially from a baseline gravity specification through incentive effects and identification strategies. Models 1–7 are estimated by Poisson Pseudo Maximum Likelihood (PPML) while Models 8–10 (which estimate country-pair effects) are estimated by Ordinary Least Squares (OLS). Models 8–10 are estimated by OLS as the computational demands of country-pair fixed effects mean that estimation

by PPML is infeasible.

All PPML models include importer, exporter, and year fixed effects throughout. The fixed effects structure uses separate importer and exporter dummies rather than the country×year approach of Baier and Bergstrand [2009], because the incentive variables of interest vary at the country-year level and would be fully absorbed by country×year fixed effects. This trade-off is discussed further in Section 4.3.

4.2.1 Model 1: Baseline Gravity

The baseline specification includes the core gravity variables with importer, exporter, and year fixed effects. Following the PPML framework, the conditional mean is specified in multiplicative form:

$$\mathbb{E}[X_{ijt}] = \exp\left(\beta_1 \ln GDP_{it} + \beta_2 \ln GDP_{jt} + \beta_3 \ln DIST_{ij} + \beta_4 REM_{it} + \beta_5 REM_{jt} + \alpha_i + \alpha_j + \delta_t\right) \tag{2}$$

where X_{ijt} is the level of bilateral flows (trade value, film count, or budget), GDP_{it} and GDP_{jt} are real GDP in constant 2015 US dollars (World Bank), $DIST_{ij}$ is population-weighted bilateral distance from the CEPII Gravity database [Conte et al., 2022], and REM_{it} is a remoteness index constructed following Baier and Bergstrand [2009] as:

$$REM_{it} = \ln\left(\frac{1}{\sum_{k \neq i} \left[GDP_{kt}/GDP_{Wt}\right]/DIST_{ik}}\right) \tag{3}$$

This index is time-varying because GDP shares evolve, even though distances are fixed. Higher values indicate greater geographic isolation from economically significant partners. The terms α_i, α_j, and δ_t denote importer, exporter, and year fixed effects respectively, and are included in all PPML specifications (Models 1–7).

4.2.2 Model 2: Full Gravity with Cultural and Institutional Controls

Model 2 augments the baseline with cultural proximity, trade policy, and institutional quality variables inside the exponential:

$$\mathbb{E}[X_{ijt}] = \exp\big(\ldots + \beta_6\, CONTIG_{ij} + \beta_7\, COMLANG_{ij} + \beta_8\, COL45_{ij} + \beta_9\, RTA_{ijt} + \beta_{10}\, EFW_{it} + \beta_{11}\, EFW_{jt}\big) \quad (4)$$

where $CONTIG_{ij}$ indicates a shared land border, $COMLANG_{ij}$ a common official language, $COL45_{ij}$ a post-1945 colonial relationship (all from CEPII), RTA_{ijt} a regional trade agreement in force, and EFW_{it} the Economic Freedom of the World composite index.

4.2.3 Model 3: Incentive Dummies

Model 3 introduces the primary variables of interest:

$$\mathbb{E}[X_{ijt}] = \exp\big(\ldots + \gamma_1\, INCENTIVE_{it}^{exp} + \gamma_2\, INCENTIVE_{jt}^{imp}\big) \quad (5)$$

where $INCENTIVE_{it}^{exp}$ equals one if the exporter (filming location) has an active incentive scheme in year t, and $INCENTIVE_{jt}^{imp}$ equals one if the importer (home country) has an active scheme.

4.2.4 Model 4: Incentive Type Heterogeneity

Model 4 disaggregates the exporter incentive by instrument type, following the three-category typology of Olsberg SPI: rebates, tax credits, and tax shelters:

$$\mathbb{E}[X_{ijt}] = \exp\big(\ldots + \gamma_1\, INCENTIVE_{it}^{exp} + \gamma_{TC}\, TAXCREDIT_{it} + \gamma_{TS}\, TAXSHELTER_{it} + \gamma_2\, INCE \quad (6)$$

Rebate is retained as the omitted reference category because it is the modal instrument in the dataset (48 of 57 countries with active schemes), providing a stable baseline against which the tax credit and tax shelter effects can be estimated. The coefficient γ_1 therefore

identifies the overall effect of an active rebate, while γ_{TC} and γ_{TS} identify the differential effect of tax credits and tax shelters respectively relative to the rebate baseline.

4.2.5 Model 5: Incentive Generosity

Model 5 replaces the binary incentive dummies with continuous generosity measures on both sides and tests whether more generous incentives generate proportionally larger effects on both the exporter and importer sides:

$$\mathbb{E}[X_{ijt}] = \exp\left(\ldots + \gamma_1 \, GENEROSITY_{it}^{exp} + \gamma_2 \, GENEROSITY_{jt}^{imp}\right) \tag{7}$$

where $GENEROSITY_{it}$ is the headline rate of the incentive scheme as sourced from the Olsberg SPI Global Incentives Index. This rate is a proportion (e.g., 0.25 for a 25% credit) and is set to zero for countries without an active scheme.

4.2.6 Models 6 and 7: Cluster Effects

Models 6 and 7 test whether incentive effects strengthen over time:

Model 6 augments Model 3 with a continuous duration variable, which would be consistent with the existance of a film industry "clustering" effect:

$$\mathbb{E}[X_{ijt}] = \exp\left(\ldots + \gamma_1 \, INCENTIVE_{it}^{exp} + \gamma_A \, YEARS_ACTIVE_{it} + \gamma_2 \, INCENTIVE_{jt}^{imp}\right) \tag{8}$$

where $YEARS_ACTIVE_{it} = \min(t - T_i^*, 20)$ when the incentive is active (T_i^* being the introduction year), capped at 20 to prevent extreme values.

Model 7 replaces the aggregate exporter incentive with separate indicators for early and late adopters:

$$\mathbb{E}[X_{ijt}] = \exp\left(\ldots + \gamma_E \, EARLY_{it} + \gamma_L \, LATE_{it} + \gamma_2 \, INCENTIVE_{jt}^{imp}\right) \tag{9}$$

where $EARLY_{it} = 1$ if country i introduced its incentive before 2005 and it is active in year t, and $LATE_{it} = 1$ for post-2005 adopters.

4.2.7 Model 8: Country-Pair Fixed Effects

Model 8 provides the most demanding identification test using country-pair fixed effects, estimated via OLS with a within-transformation. The pair fixed effect α_{ij} absorbs all time-invariant bilateral heterogeneity so only time-varying regressors survive:

$$\ln X_{ijt} = \beta_1 \ln GDP_{it} + \beta_2 \ln GDP_{jt} + \beta_3\, REM_{it} + \beta_4\, REM_{jt} + \beta_5\, RTA_{ijt}$$
$$+ \beta_6\, EFW_{it} + \beta_7\, EFW_{jt} + \gamma_1\, INCENTIVE_{it}^{exp} + \gamma_2\, INCENTIVE_{jt}^{imp} + \alpha_{ij} + \delta_t + \varepsilon_{ijt} \tag{10}$$

Estimation proceeds by demeaning all variables by their within-pair means. Singleton pairs (appearing in only one year) are dropped.

4.2.8 Model 9: Event Study

Model 9 replaces the binary exporter incentive with relative-time dummies to examine dynamic treatment effects and test for pre-existing trends:

$$\ln X_{ijt} = \boldsymbol{\beta}'\mathbf{z}_{ijt} + \gamma_2\, INCENTIVE_{jt}^{imp} + \sum_{\substack{\tau=-5 \\ \tau\neq-1}}^{+10} \mu_\tau\, D_{i\tau} + \alpha_{ij} + \delta_t + \varepsilon_{ijt} \tag{11}$$

where $D_{i\tau}$ equals one if the observation falls τ years relative to the exporter's incentive introduction date, with $\tau = -1$ as the omitted reference. The event window spans $\tau \in [-5, +10]$. The vector $\mathbf{z}_{ijt}$ includes all time-varying controls from Model 8 except the exporter incentive dummy (which is replaced by the event dummies), plus the importer incentive control.

4.2.9 Model 10: Cluster Effects with Pair Fixed Effects

Model 10 combines the years-active variable from Model 6 with the pair fixed effects identification from Model 8:

$$\ln X_{ijt} = \boldsymbol{\beta}'\mathbf{z}_{ijt} + \gamma_1 INCENTIVE_{it}^{exp} + \gamma_A YEARS_ACTIVE_{it} + \gamma_2 INCENTIVE_{jt}^{imp} + \alpha_{ij} + \delta_t + \varepsilon_{ijt} \tag{12}$$

The vector $\mathbf{z}_{ijt}$ includes all time-varying controls from Model 8 including the exporter and importer incentive dummies.

4.3 Estimator Choice and Fixed Effects Structure

Models 1–7 are estimated by Poisson Pseudo-Maximum Likelihood (PPML), following Santos Silva and Tenreyro [2006]. PPML models the conditional mean as $\mathbb{E}[X|\mathbf{z}] = \exp(\boldsymbol{\beta}'\mathbf{z})$, estimating directly on the level of the dependent variable. This offers three advantages: it accommodates the count nature of the IMDb dependent variable; it is robust to the heteroskedasticity that invalidates log-linearised OLS; and it naturally handles zero trade flows. Coefficients on log-transformed regressors (GDP, distance) are elasticities; coefficients on dummy variables (incentives, contiguity) are semi-elasticities with percentage effects given by $(\exp(\beta) - 1) \times 100\%$. Model fit is assessed via the McFadden pseudo-$R^2 = 1 - D/D_0$. All PPML models include separate importer (α_i), exporter (α_j), and year (δ_t) fixed effects to control for all time-invariant country characteristics and common temporal shocks.

Models 8–10 are estimated by OLS with country-pair fixed effects implemented via within-transformation (demeaning by pair means). The theoretically preferred country×year fixed effects of Baier and Bergstrand [2009] would fully absorb these time-varying unilateral variables, including the incentive dummies that are this paper's primary variables of interest. The separate-FE approach is standard in applied gravity work where the policy variable is unilateral rather than bilateral [e.g., Kimura and Lee, 2006].

Throughout all specifications, standard errors are clustered at the country-pair level. In panel trade data, residuals for a given country pair are likely correlated across years: if

a pair experiences an unusually productive year for film co-production, the following year is also likely to be above average due to persistent relationship-specific factors such as ongoing co-production agreements. Pair-level clustering produces standard errors are robust to both heteroskedasticity and arbitrary within-pair serial correlation [Bertrand et al., 2004, Cameron and Miller, 2015]. They are more conservative than heteroskedasticity-only robust standard errors, which assume independence and only correct for unequal variance.

4.3.1 Dependent Variables

For the IMDb dataset, all specifications are estimated with two dependent variables: the bilateral film count and the aggregate bilateral budget in constant 2010 USD. Consistency across these two measures strengthens the interpretation that incentives affect the real volume of production. The parallel estimation of each model using the BaTIS audiovisual services trade data provides independent validation of film incentives affecting broader film production flows.

4.4 Robustness and Subsample Analysis

Each pair fixed effects specification (Models 8, 9, and 10) is estimated on three samples to address potential threats to identification.

4.4.1 The Anglo-Network Problem

The English-speaking ("Anglo") production network (the United States, United Kingdom, Canada, Australia, and New Zealand) dominates global production flows and the relatively poor coverage of non-English speaking films in the IMDb data means that they are likely to be overrepresented. To account for this, each pair fixed effects specification is re-estimated on two restricted samples: (a) excluding Anglo-to-Anglo pairs only, and (b) excluding all pairs involving any Anglo country. Stability of incentive coefficients across these subsamples indicates results are not driven by idiosyncratic features of the Anglo network.

4.4.2 Sample Without Economic Freedom Controls

Because the EFW index reduces sample coverage, all pair FE models are re-estimated dropping EFW controls entirely, using the broader merged dataset. This verifies that results are not an artefact of sample selection induced by EFW data availability.

4.4.3 Income-Adjusted Economic Freedom (EFWRESID)

Following Kimura and Lee [2006], who note the correlation between economic freedom and per capita income, all ten models are additionally estimated replacing the raw EFW index with an income-adjusted variant. The residualised index ($EFWRESID$) is constructed by regressing EFW on the log of GDP in a pooled cross-sectional OLS regression across all country-year observations:

$$EFW_{it} = \alpha + \beta \cdot \ln GDP_{it} + \varepsilon_{it} \tag{13}$$

$$EFWRESID_{it} = \hat{\varepsilon}_{it} \tag{14}$$

The residual captures the component of economic freedom that is orthogonal to income level, namely institutional quality beyond what a country's GDP would predict. Stability of incentive coefficients across the raw EFW and EFWRESID specifications confirms that results are not confounded by the EFW-income correlation.

4.4.4 Robust Estimator for Staggered Adoption

The pair fixed effects specifications in Models 8 through 10 are estimated by two-way fixed effects (TWFE), which has been the standard estimator for staggered policy adoption settings. However, recent literature has shown that two-way fixed effects estimators may produce biased estimates when treatment effects vary across adoption cohorts [Borusyak et al., 2024] . To assess the robustness of the within-pair destination incentive effect to this concern, the event study specification of Model 9 is re-estimated using the imputation-based estimator of Borusyak et al. [2024]. The pair and year fixed effects are estimated using only untreated observations, and counterfactual outcomes are imputed for treated

observations using the fitted fixed effects. The average treatment effect is calculated as the mean difference between observed and imputed outcomes within the event window. Standard errors are obtained via a country-pair-clustered bootstrap with 99 replications.

The estimator is applied to the IMDb data only. Comparable application to the BaTIS audiovisual services data is not pursued: the BaTIS event study (Model 9) is already uninformative due to near-perfect multicollinearity among event-time indicators in the balanced panel structure, and re-estimation would not improve the diagnostic value of the BaTIS results on this dimension. The IMDb application is run on the same three subsamples used for the TWFE event study (full, excluding Anglo-to-Anglo pairs, and excluding all Anglo countries) and for both dependent variables (film count and budget).

5 Results

This section presents the estimation results in four stages. Section 5.1 establishes that the gravity framework fits bilateral film production flows. Section 5.2 reports the PPML incentive estimates (Models 3–7). Section 5.3 reports the fixed effects estimates (Models 8–10).. Section 5.4 compares these findings to the BaTIS robustness test results.

5.1 The Gravity Framework for Film Production

Tables 1 and 2 report the baseline and full gravity specifications for both dependent variables. The gravity model fits bilateral film flows well, with pseudo-R^2 values of 0.62–0.66, though the determinants differ from goods trade in several notable respects. Origin GDP is consistently positive and significant, though modest for film counts (elasticity ≈ 0.05) and much larger for budgets (≈ 0.6). Destination GDP is insignificant for film counts and significantly negative for budgets, inverting the standard gravity prediction. These results suggest that larger, wealthier origin countries fund more and higher-value overseas productions, but that their wealth means they have higher production costs such as wages which deter international productions.

Distance is negative and highly significant across all specifications, but the elasticity

(-0.24 to -0.28 for film counts; -0.18 for budgets) is substantially smaller than the -0.7 to -1.5 range typically found in goods trade [Head and Mayer, 2014]. This is consistent with the services trade literature, which finds that non-physical trade is less distance-sensitive [Walsh, 2006, Kimura and Lee, 2006]. The smaller distance elasticity for budgets suggests that when distant productions do occur, they tend to be higher budget.

Common language is the single most important predictor of bilateral film flows, increasing production by approximately 69% for film counts and 47% for budgets (Model 2). This is larger than the average language effect of 44% typically found in goods trade [Egger and Lassmann, 2012], which makes sense given how language-intensive films are as a product. Contiguity increases film flows by 27% for counts and 32% for budgets, less than average good trade estimate of 50% [Head and Mayer, 2014].

Colonial relationships are associated with approximately 26% *less* bilateral film production, contrary to the large positive colonial-tie effects found in goods trade. This is surprising but may reflect that post-colonial cultural differences mean that production in many ex-colonies has intentionally diverged from their colonial tradition (prominent examples include the Indian, Latin American, and Nigerian film industries). It may also reflect that film making is an industry which was relatively small in colonial times and has expanded significantly in recent decades, meaning it is less dependent on historical production networks. Regional trade agreements (RTA) are associated with 17–21% less production, less than average good trade estimate of 50% [Head and Mayer, 2014]. A potential explanation for this is that RTA partners tend to have similar economic characteristics, so co-productions are unable to take advantage of differential factors such as lower wages. This is supported by the lower contiguity estimates compared to goods trade. Finally, economic freedom is insignificant across all specifications, which contrasts with Kimura and Lee [2006] where it was found to be a strong predictor of services trade.

5.2 Incentive Effects: PPML Estimates

Tables 3–7 report the PPML incentive estimates. Adding incentive variables does not destabilise the gravity controls, which remain essentially unchanged from Model 2.

5.2.1 Aggregate Incentive Effects

Model 3 (Table 3) introduces the primary variables of interest. For film counts, the destination incentive coefficient implies that countries with incentive schemes attract 18% more international productions, while the origin coefficient implies that studios reduce overseas filming by 12% when their home country introduces incentives.

For budgets, an asymmetry emerges. The destination attraction effect is similar in magnitude (+20%) but statistically insignificant, while the origin pull-back effect is much larger (−38%) and statistically significant. Home-country incentives appear to most effectively retain high-budget productions, which have the most to gain in absolute terms from a domestic rebate on a large qualifying spend. This pattern of a strong destination effect for film counts but not budgets and a strong origin effect for budgets but not counts persists across Models 4–7.

5.2.2 Incentives: Design, Generosity, Duration and Timing

Model 4 (Table 4) disaggregates destination incentives into three instrument types: cash rebates, tax credits, and tax shelters. The specification retains the aggregate destination incentive dummy, with cash rebates as the omitted reference category; the tax credit and tax shelter coefficients capture the differential effect of those instruments relative to a rebate scheme.

For film counts, cash rebates are the most effective instrument, and the estimate is similar to the Model 3 destination coefficient. Tax credits have a +8.1% differential and tax shelters have a negative −9.3%, but these are both statistically insignificant. For budgets, cash rebates show a large positive coefficient (+30%) but it is not statistically significant. Tax credits and tax shelts have negative differentials (−8% and −67% respectively) but both are insignificant. Tax shelters having a large negative differential

is consistent with tax shelter regimes being structured around investor tax optimisation rather than genuine production attraction. However, given the small sample size (only 3 countries operate tax shelters regimes) as well as potential selection effects, we cannot draw strong conclusions from this result. This is true for tax credits too, which only cover 6 of the 57 countries with incentives and therefore doesn't constitute a robust sample.

Model 5 (Table 5) replaces dummies with continuous generosity rates. Both the destination-side and origin-side effects scale roughly linearly with the rate, with each additional 10% increase in the generosity rate for film counts and budgets roughly corresponding to a $+6 - -7\%$ destination effect and a $-7 - -15\%$ origin effect. At the mean active generosity of 27.6%, the implied destination-effect is +17% for film counts and +21% for budgets. The implied origin-effect is −13% for film counts and +21% for budgets. All estimates are statistically significant except for the budget destination-effect estimate, which is in line with the budget incentive effect also being insignificant.

Models 6 and 7 (Tables 6 and 7) test whether incentive effects accumulate over time. Model 6 finds no evidence of duration effects: each additional year of incentive operation adds approximately 0.6% to inward film production, but this is statistically insignificant. The benefit of having incentives appears to be a one-time level shift rather than a compounding advantage, providing no support to a "cluster-building" effect. However, Model 7 appears to show that *when* a country adopted incentives matters. Early adopters (pre-2005) attract 26% more films and 45% more production spending, both highly significant. Late adopters attract only 15% more films and show no statistically significant budget effect. These results appear to show that incentives do not become more effective the longer they operate, but that early adopters' incentive schemes saw success because they entered a less competitive environment.

5.3 Identification: Country-Pair Fixed Effects

Tables 8–10 report the pair fixed effects (FE) OLS estimates, which absorb all time-invariant bilateral heterogeneity and identify effects exclusively from within-pair temporal variation.

5.3.1 Pair FE Incentive Effects

Model 8 (Table 8) reports the core FE estimates. For film counts, the destination incentive effect survives at approximately +9%, half of the PPML estimate of +18%. This reduced is expected as pair FE strip out the cross-sectional correlation between incentive adoption and pre-existing production relationships. This quantifies the selection component, whereby countries with established film industries were more likely to adopt incentives, inflating the cross-sectional estimate. The portion that survives (+9%) represents the genuine within-pair effect. This result is robust across subsamples. Excluding Anglo-to-Anglo pairs yields +9.4%; excluding all Anglo countries yields +7%; dropping EFW controls yields +9.4%. The origin pull-back for film counts is −9% (compared to −12% estimate in Model 3) and significant in the full sample. The effect vanishes in the excluding-all-Anglo subsample, but the result is insignificant. Overall, both the destination and origin incentive effects are weaker in magnitude than under the PPML specifications but remain meaningful.

For budgets, the pair FE destination effect (+22%) is large and significant in both the full sample and the Anglo-to-Anglo subsamples, although it disapears when all Anglo countries are excluded. This is in contrast to the Model 3 PPML estimate, which found a similar sized but statistically insignificant effect (+20%). Meanwhile, the origin effect (−13%) is insignificant across all pair FE subsamples, which is also in contrast to the large and statistically significant PPML estimate (−38%). These results run counter to those found in Models 3–7, and overall strengthens the argument for a meaningful destination effect while undermining the case for an origin effect

5.3.2 Event Study

Model 9 (Table 9) replaces the destination incentive dummy with relative-time dummies to examine dynamic effects and test parallel trends.

For film counts, the full-sample pre-treatment coefficients are small and statistically insigniciant, with only τ_{-5} marginally significant at the 10% level. Post-treatment coefficients show a clear jump at τ_{+1} and τ_{+2}, consistent with a one-to-two year lag between

incentive introduction and production decisions responding, and estimates are consistently significant at the 5% level for the first seven years. However, when all Anglo countries are excluded, the pre-treatment coefficients are larger and individually significant, with the post-treatment average ((10%)) falling *below* the pre-treatment average ((15%)).

For budgets, the full-sample pre-treatment average is essentially zero (1%), with no individually significant coefficients. Post-treatment estimates are small initially but several estimates from τ_{+5} onward are statistically significant at the 5% and 10% level. The excluding-Anglo-to-Anglo subsample is the strongest specification: a slightly negative pre-treatment average (−10%), followed by large and increasingly significant post-treatment effects. The subsample excludes all Anglo countries shows a similar result. Overall, the film count data shows strong evidence of a causal effect and this supported to a lesser extent by the budget data.

5.3.3 Duration Under Pair FE

Model 10 (Table 10) confirms the null duration result from Model 6 under pair FE identification. The years-active coefficient is insignificant for film counts across all subsamples (0.1% − −0.4% per year) and for budgets in three of four subsamples. The one exception is the excluding-all-Anglo budget subsample, where years active is significantly *negative* (−6.9% per year). This suggests that non-Anglo destinations may see incentive effectiveness decay as competitive entry erodes their initial advantage, but the small sample size (835 observations, 72 pairs) limits this result.

5.3.4 Robust Estimator for Staggered Adoption

Table 12 11 reports the FE estimates imputation estimator of Borusyak et al. [2024], which restricts the comparison group to never-treated and not-yet-treated units and is robust to the heterogeneous-treatment-effects bias that can affect two-way FE under staggered adoption. Note that the imputation estimator does not naturally produce pre-treatment coefficients, since pre-treatment imputations are zero by construction.

For film counts, the BJS results confirms Model 8 findings. The pooled Average Treatment Effect (ATT) in the full sample is +11%, statistically significant, and close to the Model 8 estimate of +9%.The dynamic pattern is cleaner than that in Model 9: the imputed effect at τ_{+0} is essentially zero, jumps to approximately +8% at τ_{+1}, and grows to between +11% and +17% across years τ_{+2} through τ_{+8}, with most post-treatment coefficients individually significant. Both Anglo subsamples yield a smaller pooled ATT of +6% but it remains statistically significant.

For budgets, the pooled ATT is +9% in the full sample but is insignificant and there is no coherent post-treatment trajectory. The no-Anglo pair and no-Anglo subsamples yield pooled ATT estimates of −30% and −60% respectively. These are notably highly negative and reserve the sign of the Model 9 FE estimates, but only the no-Anglo estimate is statistically significant and it is dentified on a small sample (only 254 treated observations and 692 untreated observations). Nonetheless, these results suggest that weak positive budget effects in the earlier FE estimates were driven by heterogeneous-treatment-effects bias.

5.4 Cross-Dataset Validation: BaTIS Audiovisual Services

The parallel estimation on related audiovisual services data provides independent validation. The BaTIS results are reported in full in the appendix tables.

The gravity framework fits BaTIS more closely than IMDb in several respects. Both origin and destination GDP are large and highly significant, consistent with standard gravity predictions for services trade. Distance is larger and closer to goods trade estimates than the IMDb elasticities. Common language doubles audiovisual services flows (+102%), exceeding even the largest IMDb effect (+69%). Colonial ties and RTAs are both positive and significant in BaTIS, reversing the negative IMDb findings. This divergence likely reflects BaTIS capturing ongoing post-colonial media trade and the benefits of trade liberalisation for broader services flows, rather than discrete film production decisions.

The core incentive findings are similar. The destination incentive effect for BaTIS

under pair FE is +27% (compared to +45% under BaTIS PPML), which is larger than the IMDb film count effect (+9%) and also statistically significant. The BaTIS destination effect is stable across Anglo subsamples, providing stronger evidence than IMDb that the destination attraction effect generalises beyond the Anglo production network. The origin pull-back observed in IMDb PPML (−12%) is also present in BaTIS PPML (−15%), though under pair FE the BaTIS origin effect flips to *positive* (+22%) and is statistically significant. This suggests that the introduction of home-country incentives may simultaneously pull film shoots back while deepening broader audiovisual services relationships through expanded post-production and distribution activity.

BaTIS provides mixed evidence on duration effects. In PPML, years active is positive and significant (+2.2% per year) unlike the null IMDb result. However, under pair FE the coefficient reverses to significantly *negative* (−2.1% per year), indicating that the PPML duration effect reflected cross-sectional selection rather than genuine within-pair accumulation.

The Model 7 event study is uninformative due to near-perfect multicollinearity among event-time indicators in the balanced panel structure [Borusyak et al., 2024].

6 Limitations

6.1 Data Limitations

Several IMDb measurement issues affect how bilateral pairs are constructed. For films listing multiple countries of origin, the proportion of production attributable to each country is not available, while for filming destinations only the primary country is listed, so destinations that have specialised in serving as secondary locations will likely systematically undercounted. Film budgets are missing for approximately 53% of films in the sample, so the budget regressions are likely biased toward larger studio productions for which budget figures are publicly reported. The selection into the budget subsample is plausibly correlated with the production characteristics that also determine incentive responsiveness, so the budget-margin estimates should be interpreted as conditional on

this potential selection mechanism.

The most significant data limitation arises in comparing the effectiveness of different incentive types. The Olsberg SPI Global Incentives Index 2025 only provides very high-level comparable characterisations of incentive schemes, abstracting the features that determine their effectiveness in practice. The Model 4 and Model 5 estimates should be interpreted with this caveat in mind. Furthermore, subnational incentives are not captured in this analysis.

6.2 Identification and Generalisability Limitations

There are three main limitations regarding identification and validity of the results. First, the country-pair fixed effects design absorbs all time-invariant bilateral heterogeneity but cannot rule out time-varying confounders. If incentive adoption coincides with other policy or structural changes, the pair fixed effects estimator cannot separate the two. To the extent that incentive adoption is correlated with such time-varying policy changes, the pair fixed effects estimates conflate the two. Second, the sample period is dominated by the pre-streaming theatrical era. However, production location decisions in the streaming era are made by different decision-makers operating under different commercial logic, potentially limiting the applicability of these results.

Third, estimated late-adopter effects may overstate the marginal returns available to countries considering adoption today. The Model 7 finding that early adopters attracted substantially more production than late adopters likely reflects diminishing marginal returns in an increasingly crowded policy environment. However, a country considering adoption today faces an even more saturated policy landscape than the late adopters in this sample did and presumably even smaller marginal returns from introducing an incentive scheme.

7 Conclusion

This paper has examined whether national film production incentive schemes attract international film production, using a gravity model framework. I find that the gravity framework fits bilateral film production flows well but the determinants differ substantially from those documented for goods trade. I also find evidence that incentive schemes are effective at attracting international film productions (roughly $9-11\%$ under country-pair fixed effects for film counts), and weaker evidence that incentives lead to production returning home. However, when it comes to attracting film production, the timing of adoption matters more than the duration of operation, with early adopters attracting substantially more production than late adopters and no strong evidence of a cumulative "cluster effect".

Taken together, these findings indicate a genuine treatment effect rather than pure selection, but the magnitude is substantially smaller than cross-sectional estimates or reports by industry-commissioned evaluations imply. Furthermore, the incentive effect appears to be a one-time level shift attributable to entering the international subsidy competition early on, rather than a compounding advantage that grows with time and can be harnessed anytime. A country considering adoption today faces a saturated landscape in which the marginal effect of an additional scheme is plausibly smaller than even the late-adopter estimates reported here.

Two potential extensions exist. First, complementary application of the group-time average treatment effect estimator of Callaway and Sant'Anna [2021] would directly test the cohort-specific effect heterogeneity that Model 7 identifies. Second, a more comprehensive treatment of the different incentive schemes available, both by including subnational incentive schemes in the analysis and by more finely categorising the characteristics of different incentive schemes so that they can be appropriately compared to one another. These extensions would help further address how effective incentives are and identify which design elements are particularly effective, providing a more complete picture for policymakers.

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Table 1: Model 1: Baseline Gravity — IMDb (PPML)

	Film Count		Budget	
	Coef.	% effect	Coef.	% effect
ln GDP (origin)	0.0479**		0.5896***	
	(0.0222)		(0.2048)	
ln GDP (destination)	−0.0598		−0.3249**	
	(0.0677)		(0.1535)	
ln Distance	−0.2771***		−0.1795***	
	(0.0499)		(0.0449)	
Remoteness (origin)	0.2217		−1.3098***	
	(0.2723)		(0.3525)	
Remoteness (destination)	−0.1329		0.4064	
	(0.2901)		(0.7071)	
Observations	6,897		6,897	
Pseudo-R^2	0.624		0.659	

Notes: PPML estimator. Standard errors clustered by country pair in parentheses. Origin = studio home country; Destination = filming location. Coefficients on log-transformed variables are elasticities. Importer, exporter, and year fixed effects included.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 2: Model 2: Full Gravity with Cultural and Institutional Controls — IMDb (PPML)

	Film Count		Budget	
	Coef.	% effect	Coef.	% effect
ln GDP (origin)	0.0615***		0.6125**	
	(0.0234)		(0.2451)	
ln GDP (destination)	−0.0462		−0.3060**	
	(0.0534)		(0.1519)	
ln Distance	−0.2390***		−0.1761**	
	(0.0386)		(0.0699)	
Remoteness (origin)	0.2023		−1.3610***	
	(0.2654)		(0.4039)	
Remoteness (destination)	−0.1120		0.2893	
	(0.2990)		(0.7221)	
Contiguity	0.2349***	+26.5%	0.2790*	+32.2%
	(0.0670)		(0.1563)	
Common language	0.5249***	+69.0%	0.3780***	+45.9%
	(0.1397)		(0.1310)	
Colonial relationship	−0.3076**	−26.5%	−0.3499*	−29.5%
	(0.1221)		(0.2059)	
Regional trade agreement	−0.1892***	−17.2%	−0.2303*	−20.6%
	(0.0699)		(0.1395)	
Econ. freedom (dest.)	0.0062	+0.6%	0.0115	+1.2%
	(0.0405)		(0.1078)	
Econ. freedom (origin)	−0.0436	−4.3%	−0.0484	−4.7%
	(0.0571)		(0.2516)	
Observations	6,897		6,897	
Pseudo-R^2	0.661		0.663	

Notes: PPML estimator. Standard errors clustered by country pair in parentheses. Coefficients on log-transformed variables are elasticities; percentage effects for other variables computed as $(\exp(\beta) - 1) \times 100\%$. Importer, exporter, and year fixed effects included.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 3: Model 3: Incentive Dummies — IMDb (PPML)

	Film Count		Budget	
	Coef.	% effect	Coef.	% effect
ln Distance	-0.2392^{***}		-0.1767^{***}	
	(0.0384)		(0.0671)	
Contiguity	0.2370^{***}	+26.7%	0.2809^{*}	+32.4%
	(0.0674)		(0.1569)	
Common language	0.5243^{***}	+68.9%	0.3823^{***}	+46.6%
	(0.1397)		(0.1297)	
Colonial relationship	-0.3045^{**}	−26.3%	-0.3514^{*}	−29.6%
	(0.1230)		(0.2065)	
RTA	-0.1901^{***}	−17.3%	-0.2273^{*}	−20.3%
	(0.0689)		(0.1289)	
Econ. freedom (dest.)	−0.0112	−1.1%	−0.0541	−5.3%
	(0.0375)		(0.0947)	
Econ. freedom (origin)	0.0211	+2.1%	0.2688	+30.8%
	(0.0554)		(0.2412)	
Incentive (destination)	0.1687^{***}	+18.4%	0.1833	+20.1%
	(0.0511)		(0.1206)	
Incentive (origin)	-0.1267^{**}	−11.9%	-0.4747^{**}	−37.8%
	(0.0635)		(0.2381)	
Gravity controls	Yes		Yes	
Observations	6,897		6,897	
Pseudo-R^2	0.664		0.666	

Notes: PPML estimator. Standard errors clustered by country pair in parentheses. Percentage effects computed as $(\exp(\beta) - 1) \times 100\%$. GDP, remoteness, and importer/exporter/year fixed effects included but not reported.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 4: Model 4: Incentive Type Heterogeneity — IMDb (PPML)

	Film Count		Budget	
	Coef.	% effect	Coef.	% effect
ln Distance	-0.2386^{***}		-0.1772^{***}	
	(0.0383)		(0.0671)	
Common language	0.5242^{***}	+68.9%	0.3826^{***}	+46.6%
	(0.1398)		(0.1302)	
Incentive (destination)	0.1507^{***}	+16.3%	0.2631	+30.1%
	(0.0582)		(0.1672)	
Incentive (origin)	-0.1251^{*}	−11.8%	-0.4781^{**}	−38.0%
	(0.0639)		(0.2400)	
Tax credit × dest.	0.0781	+8.1%	−0.0824	−7.9%
	(0.0711)		(0.1859)	
Tax shelter × dest.	−0.0974	−9.3%	-1.0953^{***}	−66.6%
	(0.0998)		(0.2308)	
Gravity controls	Yes		Yes	
Observations	6,897		6,897	
Pseudo-R^2	0.664		0.667	

Notes: PPML estimator. The destination incentive dummy is retained from Model 3, with tax credit and tax shelter interaction dummies capturing the differential effect of those instrument types relative to the omitted reference category (cash rebate). The coefficient on "Incentive (destination)" therefore represents the effect of a cash rebate scheme. Percentage effects computed as $(\exp(\beta) - 1) \times 100\%$. Standard errors clustered by country pair. Additional gravity controls and fixed effects included but not reported.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 5: Model 5: Incentive Generosity — IMDb (PPML)

	Film Count		Budget	
	Coef.	% effect	Coef.	% effect
Generosity rate (dest.)	0.5594^{***}	+75.0%	0.6836	+98.1%
	(0.1763)		(0.4315)	
Generosity rate (origin)	-0.5121^{**}	−40.1%	-1.8463^{**}	−84.2%
	(0.2129)		(0.8015)	
Gravity controls	Yes		Yes	
Observations	6,897		6,897	
Pseudo-R^2	0.664		0.667	

Notes: PPML estimator. Generosity rate is the headline incentive rate as a proportion of qualifying expenditure (e.g., 0.25 for 25%), zero for countries without an active scheme. Percentage effects show the impact of moving from zero to the mean active generosity rate (27.6%), computed as $(\exp(\beta \times 0.276) - 1) \times 100\%$. Standard errors clustered by country pair.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 6: Model 6: Incentive Duration — IMDb (PPML)

	Film Count		Budget	
	Coef.	% effect	Coef.	% effect
Incentive (destination)	0.1529***	+16.5%	0.1656	+18.0%
	(0.0509)		(0.1365)	
Years active (dest.)	0.0062	+0.6%	0.0076	+0.8%
	(0.0064)		(0.0149)	
Incentive (origin)	−0.1203**	−11.3%	−0.4646**	−37.2%
	(0.0602)		(0.2282)	
Gravity controls	Yes		Yes	
Observations	6,897		6,897	
Pseudo-R^2	0.664		0.666	

Notes: PPML estimator. Years active = years since destination incentive introduction, capped at 20. Percentage effects computed as $(\exp(\beta)-1)\times 100\%$; for years active, this is the per-year effect. Standard errors clustered by country pair.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 7: Model 7: Early versus Late Adopters — IMDb (PPML)

	Film Count		Budget	
	Coef.	% effect	Coef.	% effect
Early adopter (dest.)	0.2301***	+25.9%	0.3740***	+45.4%
	(0.0611)		(0.1332)	
Late adopter (dest.)	0.1426**	+15.3%	0.0983	+10.3%
	(0.0591)		(0.1574)	
Incentive (origin)	−0.1255*	−11.8%	−0.4698*	−37.5%
	(0.0642)		(0.2415)	
Gravity controls	Yes		Yes	
Observations	6,897		6,897	
Pseudo-R^2	0.664		0.666	

Notes: PPML estimator. Early adopters introduced incentives before 2005; late adopters from 2005 onward. Percentage effects computed as $(\exp(\beta) - 1) \times 100\%$. Early adopter obs: 1,274; late adopter obs: 1,221. Standard errors clustered by country pair.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 8: Model 8: Country-Pair Fixed Effects — IMDb (OLS)

	Full		Excl A-to-A		Excl Anglo		No EF
	FC	Bud.	FC	Bud.	FC	Bud.	FC
ln GDP (origin)	0.0494***	0.2109***	0.0358**	0.1550**	0.0056	−0.0533	0.0460**
	(0.0191)	(0.0804)	(0.0181)	(0.0750)	(0.0151)	(0.1510)	(0.0188)
ln GDP (dest.)	−0.0072	0.0429	−0.0141	0.0441	−0.0193	−0.0829	−0.0114
	(0.0231)	(0.1474)	(0.0224)	(0.1506)	(0.0308)	(0.2967)	(0.0242)
Remoteness (origin)	0.2486	−0.6847**	0.2651*	−0.8709**	0.3771	−1.8542**	0.2431*
	(0.1516)	(0.3306)	(0.1497)	(0.3437)	(0.2304)	(0.8446)	(0.1428)
Remoteness (dest.)	−0.0256	0.4899	0.0023	0.2017	−0.1591	0.9256	−0.0176
	(0.1316)	(0.5478)	(0.1143)	(0.6111)	(0.1630)	(1.2207)	(0.1300)
Econ. freedom (dest.)	−0.0221	−0.0753	0.0113	0.0265	−0.0071	−0.0459	
	(0.0207)	(0.1165)	(0.0179)	(0.1127)	(0.0219)	(0.1799)	
Econ. freedom (origin)	−0.0434	0.2882*	0.0015	0.4437**	0.0415	0.1927	
	(0.0380)	(0.1717)	(0.0372)	(0.1759)	(0.0504)	(0.4201)	
RTA	0.0237	0.0536	−0.0262	0.0550	−0.0440	0.0737	0.0168
	(0.0347)	(0.1305)	(0.0252)	(0.1408)	(0.0287)	(0.2799)	(0.0337)
Incentive (destination)	0.0907***	0.2205**	0.0940***	0.2345**	0.0717***	−0.0003	0.0941***
	(0.0286)	(0.1020)	(0.0288)	(0.1087)	(0.0278)	(0.1669)	(0.0288)
Incentive (origin)	−0.0878***	−0.1332	−0.0989***	−0.1606	−0.0077	0.2347	−0.0915***
	(0.0261)	(0.1151)	(0.0257)	(0.1260)	(0.0244)	(0.1694)	(0.0259)
Pair FE	Yes		Yes		Yes		Yes
Year FE	Yes		Yes		Yes		Yes
Obs. (FC / Bud.)	5,995 / 2,986		5,594 / 2,638		2,733 / 835		6,092 / 2
Pairs (FC / Bud.)	920 / 684		900 / 664		556 / 357		943 / 4
Within R^2 (FC / Bud.)	.034 / .027		.031 / .025		.041 / .065		.033 / .

Notes: OLS with country-pair fixed effects. FC = ln(film count + 1); Bud. = ln(bilateral budget + 1). Standard errors clustered by country pair in parentheses. Coefficients on the log-dependent variable are approximate semi-elasticities ($\beta \times 100\% \approx$ percentage change).
"Excl A-to-A" excludes Anglo-to-Anglo pairs. "Excl Anglo" excludes all pairs involving US, UK, CA, AU, NZ. "No EFW" drops EFW controls and uses broader merged dataset.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 9: Model 9: Event Study — Destination Incentive Adoption, IMDb (Pair FE, OLS)

	Full Sample		Excl Anglo-to-Anglo		Excl All Anglo	
	FC	Bud.	FC	Bud.	FC	Bud.
$\tau - 5$	0.139*	0.040	0.141*	−0.073	0.193**	−0.274
	(0.071)	(0.239)	(0.074)	(0.256)	(0.078)	(0.695)
$\tau - 4$	0.011	−0.077	−0.001	−0.317	0.170**	−0.052
	(0.067)	(0.339)	(0.069)	(0.353)	(0.080)	(0.601)
$\tau - 3$	0.073	−0.234	0.092	−0.182	0.119*	−0.088
	(0.065)	(0.253)	(0.068)	(0.260)	(0.065)	(0.588)
$\tau - 2$	0.077	0.297	0.079	0.296	0.096	0.382
	(0.059)	(0.249)	(0.062)	(0.260)	(0.066)	(0.587)
$\tau - 1$	ref.	ref.	ref.	ref.	ref.	ref.
$\tau + 0$	0.065	0.191	0.062	0.237	0.080	0.184
	(0.058)	(0.198)	(0.060)	(0.217)	(0.080)	(0.346)
$\tau + 1$	0.168***	0.120	0.162***	0.150	0.206***	0.223
	(0.056)	(0.236)	(0.058)	(0.250)	(0.066)	(0.415)
$\tau + 2$	0.172***	0.073	0.163**	0.130	0.211***	0.357
	(0.064)	(0.271)	(0.068)	(0.294)	(0.076)	(0.535)
$\tau + 3$	0.091	0.393	0.076	0.501	0.084	1.237**
	(0.069)	(0.291)	(0.072)	(0.307)	(0.080)	(0.605)
$\tau + 4$	0.161**	0.251	0.135*	0.344	0.088	0.455
	(0.074)	(0.264)	(0.080)	(0.286)	(0.092)	(0.596)
$\tau + 5$	0.153**	0.462*	0.148*	0.580**	0.112	1.290**
	(0.074)	(0.240)	(0.081)	(0.261)	(0.076)	(0.609)
$\tau + 6$	0.210**	0.510**	0.190**	0.726***	0.130	0.789
	(0.085)	(0.249)	(0.091)	(0.267)	(0.087)	(0.676)
$\tau + 7$	0.194**	0.394	0.142	0.591*	0.130	0.995
	(0.094)	(0.322)	(0.104)	(0.357)	(0.125)	(0.769)
$\tau + 8$	0.150	0.647*	0.156	1.067***	0.072	2.094**
	(0.099)	(0.358)	(0.108)	(0.352)	(0.107)	(0.951)
$\tau + 9$	0.159	0.537*	0.114	0.909***	0.101	1.869**
	(0.102)	(0.297)	(0.113)	(0.309)	(0.128)	(0.817)
$\tau + 10$	0.181*	0.839**	0.132	0.944**	0.021	1.185
	(0.106)	(0.359)	(0.116)	(0.413)	(0.103)	(1.543)
Incentive (origin)	0.022	0.069	0.014	−0.000	0.055	0.270
	(0.053)	(0.239)	(0.055)	(0.264)	(0.060)	(0.306)
Avg. pre (τ_{-5} to τ_{-2})	0.075	0.007	0.078	−0.069	0.145	−0.008
Avg. post (τ_0 to τ_{+10})	0.155	0.402	0.135	0.562	0.112	0.971
Obs. (FC / Bud.)	1,735 / 821		1,606 / 712		821 / 223	
Pairs (FC / Bud.)	350 / 176		336 / 163		202 / 71	

Notes: Event study coefficients relative to τ_{-1} (year before destination incentive adoption). OLS with country-pair FE. FC = ln(film count + 1); Bud. = ln(bilateral budget + 1). Standard errors clustered by country pair in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 10: Model 10: Incentive Duration with Country-Pair Fixed Effects — IMDb (OLS)

	Full		Excl A-to-A		Excl Anglo		No EFW	
	FC	Bud.	FC	Bud.	FC	Bud.	FC	B
Incentive (destination)	0.0792***	0.2120**	0.0850***	0.2204*	0.0694**	0.0889	0.0806***	0
	(0.0290)	(0.1073)	(0.0291)	(0.1128)	(0.0283)	(0.1680)	(0.0292)	(0
Years active (destination)	0.0038	0.0031	0.0032	0.0060	0.0008	−0.0688***	0.0043	0
	(0.0029)	(0.0103)	(0.0030)	(0.0118)	(0.0032)	(0.0222)	(0.0028)	(0
Incentive (origin)	−0.0878***	−0.1343	−0.0993***	−0.1638	−0.0081	0.3164*	−0.0913***	−0
	(0.0262)	(0.1154)	(0.0259)	(0.1262)	(0.0246)	(0.1671)	(0.0261)	(0
Gravity controls	Yes		Yes		Yes		Yes	
Pair FE	Yes		Yes		Yes		Yes	
Year FE	Yes		Yes		Yes		Yes	
Obs. (FC / Bud.)	5,995 / 2,986		5,594 / 2,638		2,733 / 835		6,092 / 2,77	
Within R^2 (FC / Bud.)	.035 / .027		.031 / .025		.041 / .085		.034 / .027	

Notes: OLS with country-pair FE. FC = ln(film count + 1); Bud. = ln(bilateral budget + 1). Years active = years since destination incentive introduction, capped at 20. Standard errors clustered by country pair in parentheses. Coefficients on the log-dependent variable are approximate semi-elasticities ($\beta \times 100\% \approx$ percentage change). Gravity controls (GDP, remoteness, RTA, EFW where applicable) included but not reported.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 11: BJS Imputation Estimator — Robustness Check on Model 9 Event Study, IMDb

	Full Sample		Excl Anglo-to-Anglo		Excl All Anglo	
	FC	Bud.	FC	Bud.	FC	Bud.
$\tau-5$ to $\tau-2$	*Not estimated by construction (see notes)*					
$\tau-1$	ref.	ref.	ref.	ref.	ref.	ref.
$\tau+0$	0.014	0.102	0.009	0.298*	0.036	0.301
	(0.040)	(0.187)	(0.032)	(0.178)	(0.045)	(0.271)
$\tau+1$	0.079**	−0.076	0.106***	−0.074	0.094**	−0.204
	(0.038)	(0.149)	(0.039)	(0.180)	(0.044)	(0.309)
$\tau+2$	0.112**	−0.294	0.080*	−0.646***	0.104**	−1.018**
	(0.045)	(0.225)	(0.043)	(0.226)	(0.043)	(0.446)
$\tau+3$	0.114**	0.360*	0.047	−0.041	0.038	−0.233
	(0.049)	(0.218)	(0.041)	(0.329)	(0.048)	(0.521)
$\tau+4$	0.127***	0.392**	0.064	−0.111	0.017	−0.223
	(0.049)	(0.174)	(0.041)	(0.246)	(0.037)	(0.423)
$\tau+5$	0.141***	0.276	0.089**	−0.514	0.062	−0.635
	(0.044)	(0.215)	(0.042)	(0.338)	(0.044)	(0.623)
$\tau+6$	0.124**	0.075	0.057	−0.301	0.031	−1.043*
	(0.052)	(0.288)	(0.045)	(0.328)	(0.039)	(0.557)
$\tau+7$	0.158***	0.016	0.076	−0.708**	0.075	−1.459***
	(0.057)	(0.295)	(0.055)	(0.301)	(0.081)	(0.550)
$\tau+8$	0.166***	0.401*	0.095**	−0.157	0.097*	−0.695
	(0.053)	(0.240)	(0.048)	(0.302)	(0.052)	(0.590)
$\tau+9$	0.036	−0.223	0.005	−0.591	0.052	−0.704
	(0.059)	(0.228)	(0.058)	(0.406)	(0.054)	(0.619)
$\tau+10$	0.135**	0.153	0.029	−0.790*	0.008	−1.548*
	(0.064)	(0.251)	(0.054)	(0.417)	(0.045)	(0.860)
Pooled ATT	**0.107*** **	**0.087**	**0.061*** **	**−0.290***	**0.056** **	**−0.580** **
	(0.032)	(0.123)	(0.024)	(0.151)	(0.026)	(0.257)
TWFE comparison (Model 8)	0.091***	0.221**	0.094***	0.235**	0.072***	−0.000
	(0.029)	(0.102)	(0.029)	(0.109)	(0.028)	(0.167)
Treated obs. in event window	1,470	719	1,183	528	751	254
Untreated obs. in pool	4,402 / 1,958		3,713 / 1,482		2,401 / 692	
Bootstrap reps	99		99		99	

Notes: Imputation estimator of Borusyak et al. [2024] applied to the Model 9 event study specification with country-pair fixed effects. The estimator fits pair and year fixed effects on untreated observations only (those for which the destination has not yet adopted an incentive scheme), then imputes counterfactual outcomes for treated observations and computes ATTs as the average difference between observed and imputed outcomes. By construction, pre-treatment ATTs are zero and are not separately estimated; pre-trend tests must be conducted via the TWFE event study reported in Table 9. Standard errors are obtained via country-pair-clustered bootstrap with 99 replications and reported in parentheses. The pooled ATT is the average of post-treatment ATTs weighted by the number of treated observations at each event time.
FC = ln(film count + 1); Bud. = ln(bilateral budget + 1). Untreated obs. in pool reports counts for FC / Bud. separately due to the smaller sample available for the budget specification (see Section 6.1). The TWFE comparison row reproduces the destination incentive coefficient from Model 8 (Table 8) for direct comparison; sample sizes differ between BJS and TWFE because BJS conditions on observations within the $[-5, +10]$ event window while Model 8 uses the full panel.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 12: Model 1: Baseline Gravity — BaTIS Audiovisual Services (PPML)

	Coef.	% effect
ln GDP (origin)	0.9223***	
	(0.2551)	
ln GDP (destination)	0.7022***	
	(0.2168)	
ln Distance	−0.6995***	
	(0.0411)	
Remoteness (origin)	0.4266	
	(0.6685)	
Remoteness (destination)	−2.3460***	
	(0.8035)	
Observations	320,934	

Notes: PPML estimator. Dependent variable: bilateral audiovisual services exports (BaTIS, constant 2010 USD). Standard errors clustered by country pair in parentheses. Coefficients on log-transformed variables are elasticities. Origin = studio home country; Destination = filming location. Importer, exporter, and year fixed effects included.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 13: Model 2: Full Gravity with Cultural and Institutional Controls — BaTIS (PPML)

	Coef.	% effect
ln GDP (origin)	1.0467***	
	(0.2659)	
ln GDP (destination)	0.6997***	
	(0.2194)	
ln Distance	−0.5230***	
	(0.0408)	
Remoteness (origin)	0.4215	
	(0.6529)	
Remoteness (destination)	−2.1867***	
	(0.7756)	
Contiguity	0.1780*	+19.5%
	(0.1063)	
Common language	0.7006***	+101.5%
	(0.0787)	
Colonial relationship	0.4537***	+57.4%
	(0.1630)	
Regional trade agreement	0.2445***	+27.7%
	(0.0792)	
Econ. freedom (dest.)	0.0025	+0.3%
	(0.1003)	
Econ. freedom (origin)	−0.2500***	−22.1%
	(0.0753)	
Observations	320,934	

Notes: PPML estimator. Standard errors clustered by country pair in parentheses. Coefficients on log-transformed variables are elasticities; percentage effects for other variables computed as $(\exp(\beta) - 1) \times 100\%$. Importer, exporter, and year fixed effects included.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 14: Model 3: Incentive Dummies — BaTIS (PPML)

	Coef.	% effect
ln Distance	-0.5247^{***}	
	(0.0408)	
Contiguity	0.1783^{*}	+19.5%
	(0.1064)	
Common language	0.7029^{***}	+102.0%
	(0.0787)	
Colonial relationship	0.4534^{***}	+57.4%
	(0.1635)	
RTA	0.2371^{***}	+26.8%
	(0.0783)	
Econ. freedom (dest.)	−0.0499	−4.9%
	(0.0913)	
Econ. freedom (origin)	-0.2294^{***}	−20.5%
	(0.0713)	
Incentive (destination)	0.3761^{***}	+45.7%
	(0.0895)	
Incentive (origin)	-0.1640^{**}	−15.1%
	(0.0742)	
Gravity controls	Yes	
Observations	320,934	

Notes: PPML estimator. Percentage effects computed as $(\exp(\beta) - 1) \times 100\%$. GDP, remoteness, and importer/exporter/year fixed effects included but not reported.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 15: Model 4: Incentive Type Heterogeneity — BaTIS (PPML)

	Coef.	% effect
ln Distance	-0.5248^{***}	
	(0.0408)	
Common language	0.7029^{***}	+102.0%
	(0.0787)	
Incentive (destination)	0.3871^{***}	+47.3%
	(0.0921)	
Incentive (origin)	-0.1646^{**}	−15.2%
	(0.0742)	
Tax credit × dest.	−0.3395	−28.8%
	(0.2096)	
Tax shelter × dest.	−0.0186	−1.8%
	(0.1695)	
Gravity controls	Yes	
Observations	320,934	

Notes: PPML estimator. The destination incentive dummy is retained from Model 3, with tax credit and tax shelter interaction dummies capturing the differential effect of those instrument types relative to the omitted reference category (cash rebate). The coefficient on "Incentive (destination)" therefore represents the effect of a cash rebate scheme. Percentage effects computed as $(\exp(\beta) - 1) \times 100\%$. Standard errors clustered by country pair. Additional gravity controls and fixed effects included but not reported.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 16: Model 5: Incentive Generosity — BaTIS (PPML)

	Coef.	% effect
Generosity rate (dest.)	1.0941^{***}	+198.6%
	(0.2991)	
Generosity rate (origin)	-0.5507^{**}	−42.3%
	(0.2381)	
Gravity controls	Yes	
Observations	320,934	

Notes: PPML estimator. Generosity rate is the headline incentive rate as a proportion of qualifying expenditure, zero for countries without an active scheme. Mean generosity among active destinations: 27.8%. Percentage effects show the full $(\exp(\beta) - 1) \times 100\%$ transformation. Standard errors clustered by country pair.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 17: Model 6: Incentive Duration — BaTIS (PPML)

	Coef.	% effect
Incentive (destination)	0.3403^{***}	+40.5%
	(0.0855)	
Years active (dest.)	0.0218^{**}	+2.2%
	(0.0092)	
Incentive (origin)	-0.1548^{**}	−14.3%
	(0.0723)	
Gravity controls	Yes	
Observations	320,934	

Notes: PPML estimator. Years active = years since destination incentive introduction, capped at 20. Mean years active (when >0): 8.3, median: 7.0. Percentage effects computed as $(\exp(\beta) - 1) \times 100\%$; for years active, this is the per-year effect. Standard errors clustered by country pair.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 18: Model 7: Early versus Late Adopters — BaTIS (PPML)

	Coef.	% effect
Early adopter (dest.)	dropped (collinear)	
Late adopter (dest.)	0.3761^{***}	+45.7%
	(0.0895)	
Incentive (origin)	-0.1640^{**}	−15.1%
	(0.0742)	
Gravity controls	Yes	
Observations	320,934	

Notes: PPML estimator. Early adopter dummy dropped due to multicollinearity with the exporter fixed effects (BaTIS coverage begins 2005, by which point all early adopters already had incentives in place). The late adopter coefficient is therefore identified relative to countries that never adopted incentives. Standard errors clustered by country pair.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 19: Model 8: Country-Pair Fixed Effects — BaTIS (OLS)

	Full	Excl A-to-A	Excl Anglo	No EFW
ln GDP (origin)	0.9854***	0.9853***	0.9987***	0.9955***
	(0.0472)	(0.0472)	(0.0495)	(0.0412)
ln GDP (dest.)	0.4302***	0.4303***	0.4197***	0.4084***
	(0.0425)	(0.0425)	(0.0440)	(0.0378)
Remoteness (origin)	−0.0814	−0.0794	0.0159	0.0002
	(0.1615)	(0.1616)	(0.1714)	(0.1490)
Remoteness (dest.)	−0.6366***	−0.6370***	−0.5103***	−0.6326***
	(0.1858)	(0.1859)	(0.1960)	(0.1683)
Econ. freedom (dest.)	0.0054	0.0054	−0.0070	
	(0.0216)	(0.0216)	(0.0223)	
Econ. freedom (origin)	0.1012***	0.1012***	0.1270***	
	(0.0210)	(0.0210)	(0.0221)	
RTA	−0.0979***	−0.0982***	−0.1011**	−0.1087***
	(0.0369)	(0.0370)	(0.0395)	(0.0344)
Incentive (destination)	0.2673***	0.2674***	0.2568***	0.2383***
	(0.0244)	(0.0244)	(0.0256)	(0.0224)
Incentive (origin)	0.2239***	0.2238***	0.2565***	0.2356***
	(0.0245)	(0.0245)	(0.0258)	(0.0228)
Pair FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	320,166	319,791	293,851	398,431
Pairs	20,934	20,914	19,445	26,883
Within R^2	0.015	0.015	0.015	0.014

Notes: OLS with country-pair fixed effects. Dependent variable: ln(bilateral audiovisual services exports + 1). Standard errors clustered by country pair in parentheses. Coefficients on the log-dependent variable are approximate semi-elasticities ($\beta \times 100\% \approx$ percentage change).
"Excl A-to-A" excludes Anglo-to-Anglo pairs. "Excl Anglo" excludes all pairs involving US, UK, CA, AU, NZ. "No EFW" drops EFW controls and uses broader merged dataset.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 20: Model 9: Event Study — Destination Incentive Adoption, BaTIS (Pair FE, OLS)

	Full	Excl A-to-A	Excl Anglo	No EFW
$\tau - 5$	−0.225	−0.228	−0.173	−0.207
$\tau - 4$	−0.121	−0.123	−0.084	−0.105
$\tau - 3$	−0.092	−0.093	−0.066	−0.087
$\tau - 2$	−0.099	−0.100	−0.077	−0.099
$\tau - 1$	ref.	ref.	ref.	ref.
$\tau + 0$	0.011	0.011	−0.012	0.003
$\tau + 1$	0.007	0.008	−0.045	0.005
$\tau + 2$	0.018	0.021	−0.018	0.021
$\tau + 3$	0.070	0.074	−0.021	0.072
$\tau + 4$	−0.049	−0.045	−0.169	−0.045
$\tau + 5$	−0.018	−0.013	−0.154	−0.021
$\tau + 6$	−0.034	−0.028	−0.187	−0.028
$\tau + 7$	−0.056	−0.050	−0.240	−0.043
$\tau + 8$	−0.102	−0.095	−0.290	−0.070
$\tau + 9$	−0.080	−0.072	−0.304	−0.050
$\tau + 10$	−0.077	−0.068	−0.325	−0.027
Incentive (origin)	0.156***	0.155***	0.177***	0.177***
	(0.042)	(0.042)	(0.043)	(0.041)
Avg. pre (τ_{-5} to τ_{-2})	−0.134	−0.136	−0.100	−0.124
Avg. post (τ_0 to τ_{+10})	−0.028	−0.023	−0.161	−0.017
Observations	80,514	80,416	74,304	91,381
Pairs	7,732	7,720	7,065	8,846

Notes: Event study coefficients relative to τ_{-1}. OLS on ln(BaTIS exports + 1) with country-pair FE. Standard errors on individual event-time dummies are extremely large (order of thousands) due to near-perfect multicollinearity among event-time indicators in balanced panel data; no individual coefficient is statistically significant. The origin incentive coefficient remains precisely estimated. Coefficients are reported for completeness but individual significance should not be interpreted.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 21: Model 10: Incentive Duration with Country-Pair Fixed Effects — BaTIS (OLS)

	Full	Excl A-to-A	Excl Anglo	No EFW
Incentive (destination)	0.3051***	0.3053***	0.3025***	0.2802***
	(0.0245)	(0.0246)	(0.0259)	(0.0226)
Years active (destination)	−0.0208***	−0.0209***	−0.0242***	−0.0214***
	(0.0036)	(0.0036)	(0.0041)	(0.0033)
Incentive (origin)	0.2221***	0.2221***	0.2549***	0.2334***
	(0.0245)	(0.0245)	(0.0258)	(0.0228)
Gravity controls	Yes	Yes	Yes	Yes
Pair FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	320,166	319,791	293,851	398,431
Within R^2	0.016	0.016	0.016	0.014

Notes: OLS with country-pair FE. Dependent variable: ln(BaTIS audiovisual services exports + 1). Years active = years since destination incentive introduction, capped at 20. Standard errors clustered by country pair in parentheses. Coefficients on the log-dependent variable are approximate semi-elasticities ($\beta \times 100\% \approx$ percentage change). Gravity controls (GDP, remoteness, RTA, EFW where applicable) included but not reported.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.